Cheng Yuan (Ross) King

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CAREER ASPIRATION

I am a software engineer and MSc Artificial Intelligence candidate specialising in the testing and reliability of AI systems. I design and release open-source software that automatically tests AI and machine-learning models for failures before they reach users, and data platforms that keep large, inconsistent datasets accurate. I am skilled in Python and SQL, automated testing and continuous integration, and building and deploying services with FastAPI, Docker and cloud data tools.

EDUCATION

University of Sheffield Sep 2025 – Sep 2026 | Sheffield, UK
MSc Artificial Intelligence

University of Sheffield International College Jan 2025 – Aug 2025 | Sheffield, UK
Pre-Master's Computing and Computer Science

Queen's University Belfast Sep 2021 – Jun 2024 | Belfast, UK
BSc Computer Science

SKILLS

Languages : Python, SQL, Bash, Git
AI & Evaluation : LLM evaluation, AISI Inspect, red-teaming, prompt-injection testing, RAG evaluation, agent tooling (MCP)
Testing & Release : pytest, CI/CD, GitHub Actions, Docker, mypy (strict), PyPI packaging
Services & APIs : FastAPI, REST/JSON APIs
Data : dbt, PySpark, Airflow, BigQuery
Spoken : English (Fluent), Mandarin Chinese (Native), Japanese (JLPT N1)

PROJECTS

Agent Release Gates | *Python, Inspect, FastAPI, Docker, CI*

- Built a release gate for LLM agents that replays **8** recorded incidents against any changed agent, prompt, model or tool policy and returns `ship` / `warn` / `block` — so an engineering team catches a behavioural regression in CI instead of after users hit it.
- Found and published the failure that mattered: the retriever scores **99.31%** on the project's own benchmark but **79.92%** on **640** external public cases (TechQA, WixQA; 510 documents). The in-house benchmark is circular by construction, so the external number is the one reported.
- Shipped end to end — CLI on PyPI, Inspect (UK AISI) evaluation suite, FastAPI service, Docker, CI and **305** tests — with a reviewer dashboard as the single evidence surface.

Red-Team Foundry | *Python, Inspect, PyPI*

- Tested whether the public jailbreak benchmarks the field still cites can actually tell models apart: ran **1,883** prompts from **4** pinned adversarial corpora across **12** evaluation cells (2 target models $\times$ 2 attack families $\times$ up to 4 defence stacks), with bootstrap confidence intervals and per-run API cost.
- Attack success came out at **0–4%** — a negative result — so proved the harness was not simply under-testing: a known-vulnerable model through the identical pipeline scored **80%** attack success, with two independent LLM judges agreeing at κ = **+0.935**.
- Reported where the agreement metric collapses instead of banking it — in 11 of 12 cells both judges labelled every case identically, which makes κ undefined rather than perfect — and wrote a CI check that fails the build if that caveat is dropped. **340** tests; published to PyPI; exports to Inspect logs.

TfL Cycle-Hire Data Platform | *PySpark, dbt, DuckDB, Airflow, MCP*

- Unified **41.4M** London cycle-hire journeys from **148** files across **5** incompatible schema eras into one modelled warehouse, so an analyst queries a decade of history as a single table instead of reconciling formats by hand.
- Carried **92** data tests over **17** dbt models, so an upstream schema change breaks the build rather than silently corrupting a published figure.
- Measured disruption impact at **1.42**$\times$ median station-day demand, reported as association, not causation, and exposed the warehouse through a read-only MCP server so an LLM agent can query it without credentials.